

SQL Projects on Consumer Goods

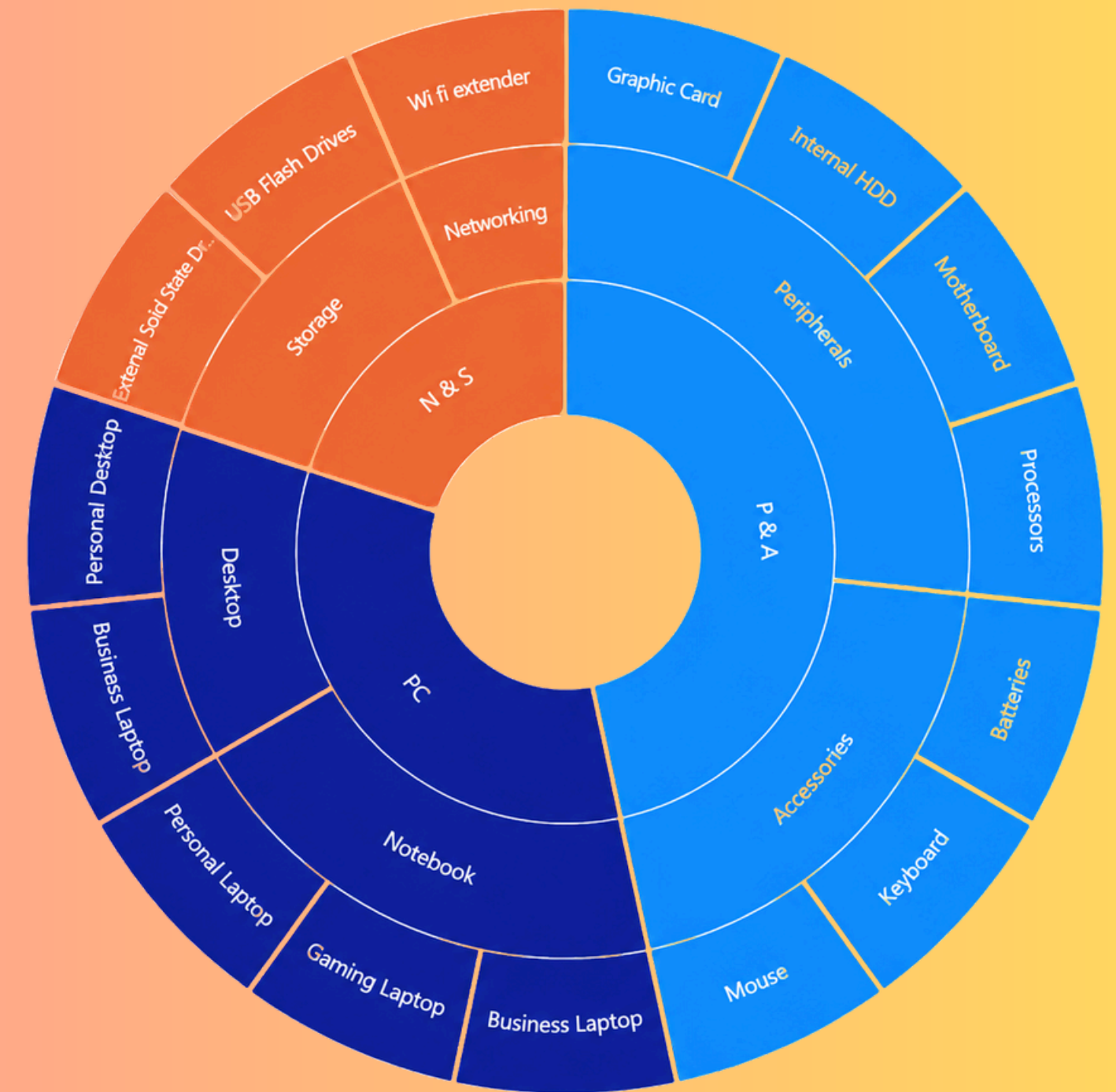


Presented by

S K Adityanarayan

Company Overview

- AtliQ Hardware is a leading computer hardware manufacturer in India, known for delivering high-quality and reliable products.
- The company has established a strong domestic presence and continues to expand its footprint across international markets.
- With a focus on innovation and customer-centric solutions, AtliQ Hardware supports diverse business needs in a rapidly evolving tech landscape.
- AtliQ is following September – August Financial Year.



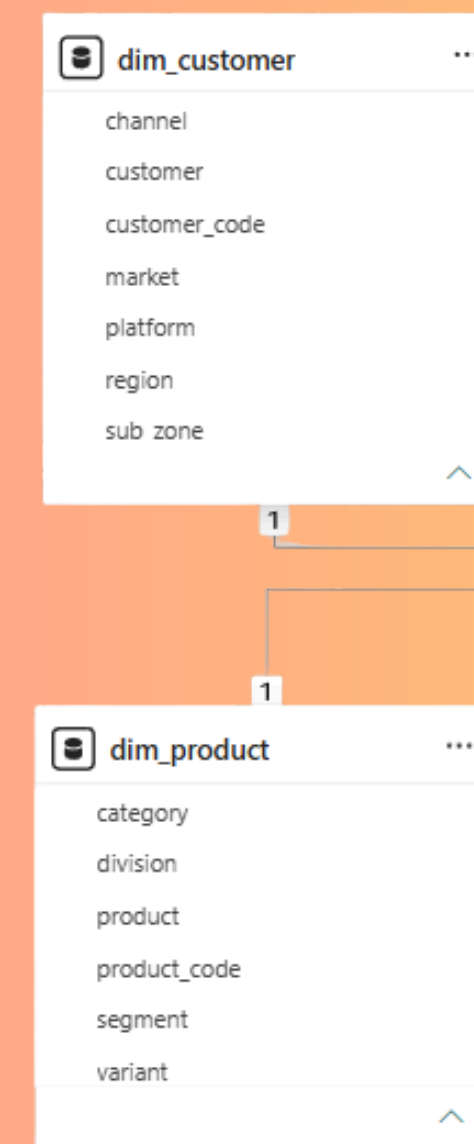
Problem Statement

- Rapid regional expansion and increasing transaction volumes have made it difficult for management to derive timely, data-driven insights.
- To address this gap, the company plans to build a dedicated data analytics team under the leadership of Tony Sharma, Senior Data Analyst.
- A structured SQL challenge is introduced to identify candidates with strong technical expertise as well as effective communication and problem-solving skills.
- The challenge includes 10 real-world business requirements that candidates must solve using SQL to demonstrate their analytical capabilities.

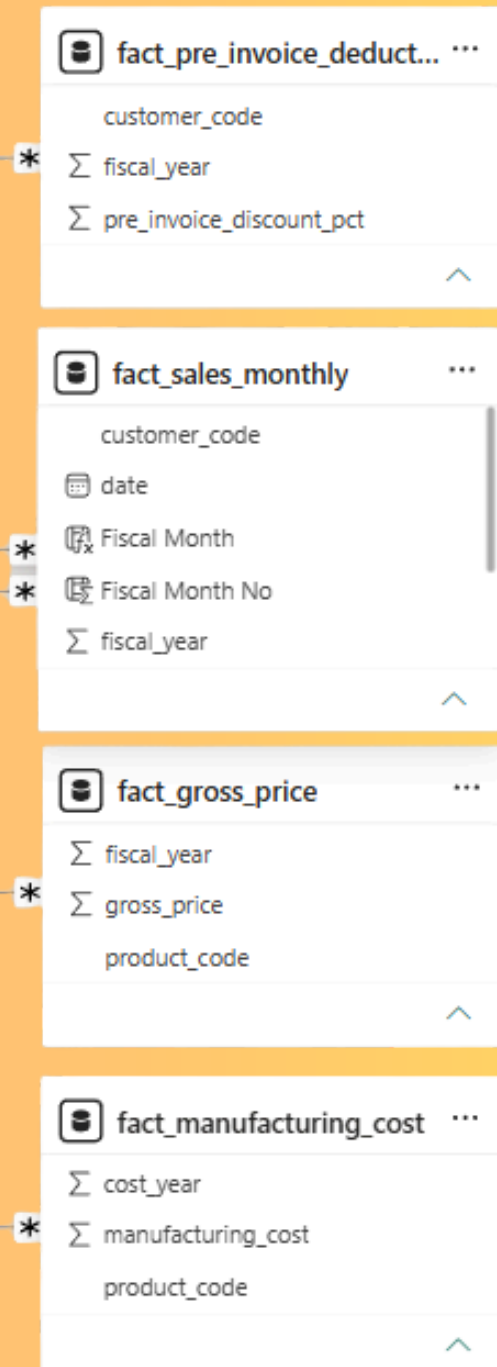
Data Model — AtliQ Hardware

- This project uses a star schema with 6 tables — 2 dimension tables (Customer, Product) and 4 fact tables (Sales, Gross Price, Manufacturing Cost, Pre-Invoice Deductions).
- All fact tables join to dimensions via customer_code or product_code, enabling cross-dimensional analysis.
- Tool Used : MySQL | Power BI (Visualization) | Presentation: Canva

Dimension Tables



Fact Tables



Ad Hoc Request - 1

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

```
SELECT
  DISTINCT market
FROM dim_customer
WHERE customer = 'Atliq Exclusive'
      AND region = 'APAC'
ORDER BY market;
```

- India
- Bangladesh
- South Korea
- Japan
- Indonesia
- Australia

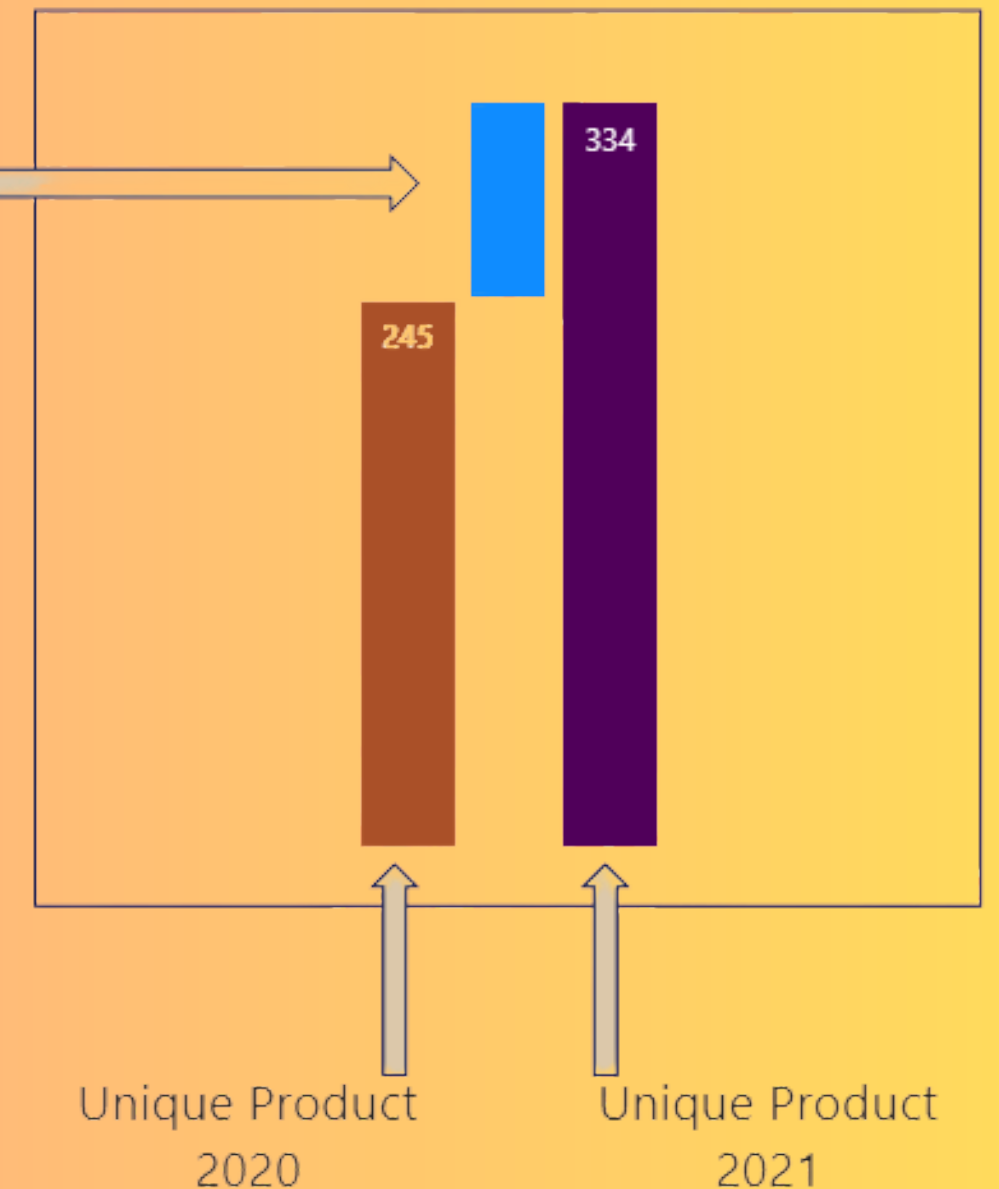


Ad Hoc Request - 2

What is the percentage of unique product increase in 2021 vs. 2020?

```
10 • WITH product_count AS (  
11     SELECT  
12         fiscal_year,  
13         COUNT(DISTINCT product_code) AS unique_product  
14     FROM fact_sales_monthly  
15     WHERE fiscal_year IN (2020,2021)  
16     GROUP BY fiscal_year  
17 )  
18 SELECT  
19     MAX(CASE WHEN fiscal_year = 2020 THEN unique_product END) AS unique_product_2020,  
20     MAX(CASE WHEN fiscal_year = 2021 THEN unique_product END) AS unique_product_2021,  
21     ROUND(  
22         (  
23             MAX(CASE WHEN fiscal_year = 2021 THEN unique_product END) -  
24             MAX(CASE WHEN fiscal_year = 2020 THEN unique_product END)  
25         ) * 100 / MAX(CASE WHEN fiscal_year = 2020 THEN unique_product END), 2) AS percentage_chg  
26 FROM product_count;
```

Unique Product
increased by **36.33 %**



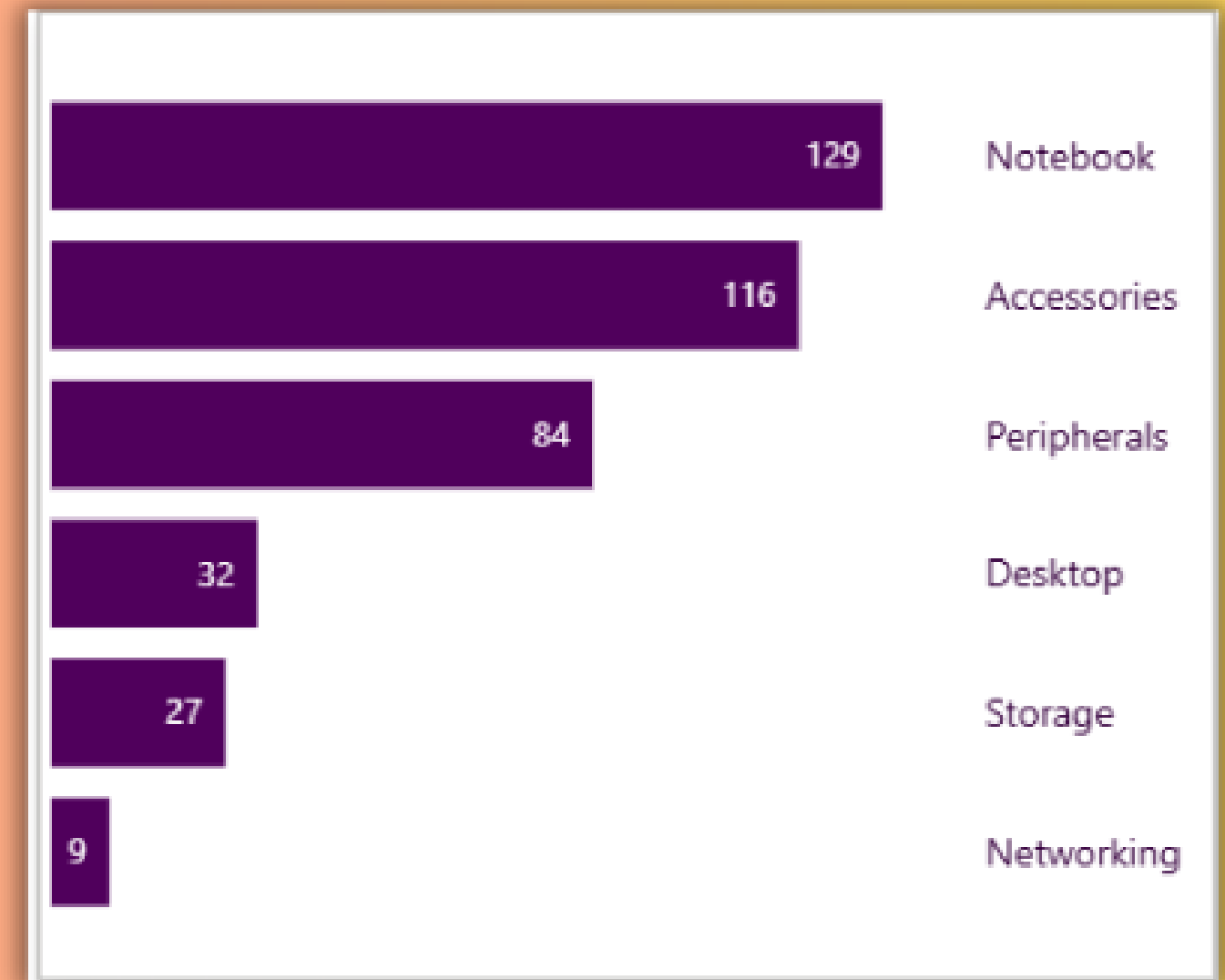
Insights

Unique product count grew by **36.33%** in FY2021 compared to FY2020, rising from 245 to 334 products.

Ad Hoc Request - 3

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts

```
SELECT
    segment,
    COUNT(product_code) AS product_count
FROM dim_product
GROUP BY segment
ORDER BY product_count DESC;
```



Insights

- Notebook, Accessories, and Peripherals collectively account for ~83% of total unique products (329 out of 397), making them the dominant segments in the product portfolio.

Ad Hoc Request - 3



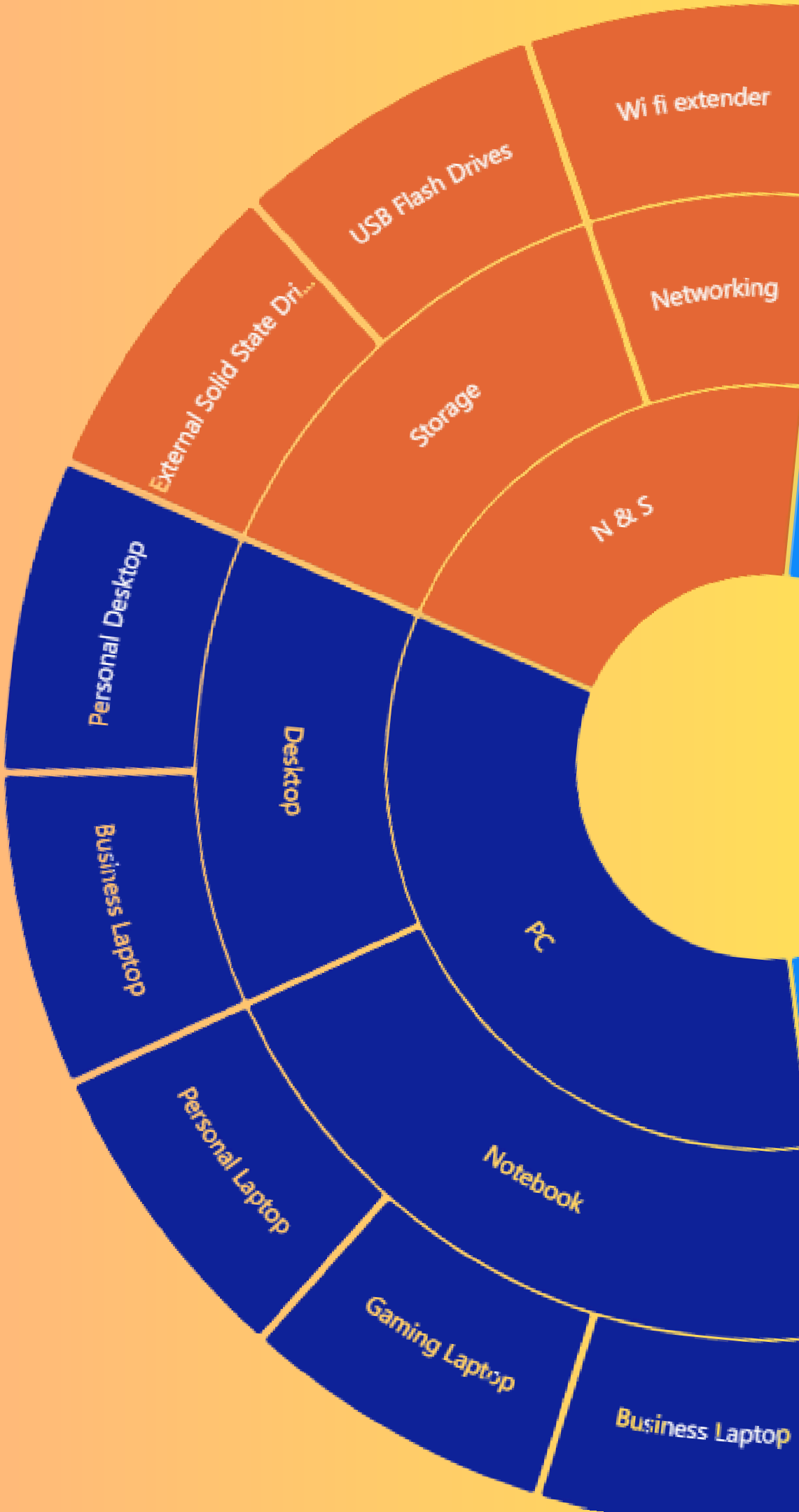
segment	Avg GS	Avg MC	Gross Mirgin
Peripherals	1,814.41	540.92	70.19%
Accessories	1,816.42	543.74	70.07%
Notebook	2,255.44	674.68	70.09%
Desktop	2,553.72	767.44	69.95%
Storage	2,986.82	897.70	69.94%
Networking	4,953.04	1,473.32	70.25%

Avg GS : Average Gross Sale
Avg MC : Average Manufacturing Cost

- External Solid State Drive
- USB Flash Drives
- Wifi Extender
- Personal Laptop / Desktop
- Business Laptop / Desktop

Suggestions

- Package Deal
- Gift Vouchers
- Card Discounts
- Customer Services
- Students Discounts
- Loyalty Points
- Membership



Ad Hoc Request - 4

Which segment had the most increase in unique products in 2021 vs 2020

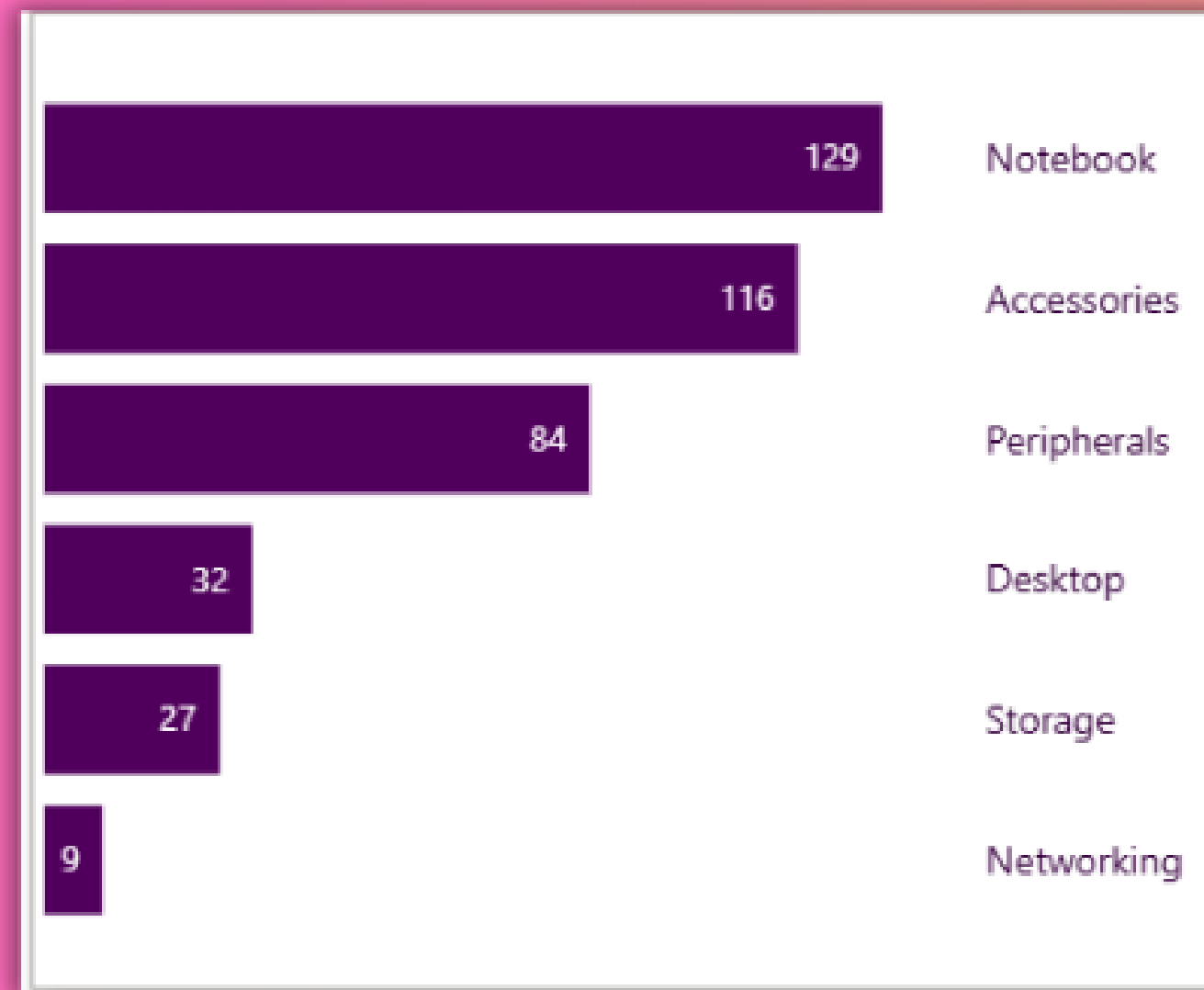
```
WITH product_count AS (  
  SELECT  
    p.segment,  
    f.fiscal_year,  
    COUNT(DISTINCT f.product_code) AS unique_product  
  FROM fact_sales_monthly f  
  JOIN dim_product p  
  ON f.product_code = p.product_code  
  WHERE fiscal_year IN (2020,2021)  
  GROUP BY p.segment,f.fiscal_year  
)  
SELECT  
  segment,  
  MAX(CASE WHEN fiscal_year = 2020 THEN unique_product END) AS unique_product_2020,  
  MAX(CASE WHEN fiscal_year = 2021 THEN unique_product END) AS unique_product_2021,  
  (  
    MAX(CASE WHEN fiscal_year = 2021 THEN unique_product END) -  
    MAX(CASE WHEN fiscal_year = 2020 THEN unique_product END)  
  ) AS difference  
FROM product_count  
GROUP BY segment  
ORDER BY difference DESC;
```

Segment	2020	2021	Increased ▼
Accessories	69	103	34 ▲
Notebook	92	108	16 ▲
Peripherals	59	75	16 ▲
Desktop	7	22	15 ▲
Storage	12	17	5 ▲
Networking	6	9	3 ▲

Insights

- Accessories segment introduced **Highest 34** new Unique products
- Networking segment is the least performing of 3 new Unique products.

Desktop Segment Performance Analysis (2020–2021)



Segment	2020	2021	Increased %
Desktop	7	22	214.29%
Networking	6	9	50.00%
Accessories	69	103	49.28%
Storage	12	17	41.67%
Peripherals	59	75	27.12%
Notebook	92	108	17.39%

Insights

- In FY 2021 Desktop segment reported strong 214% move in Unique Product Counts.

Ad Hoc Request - 5

Get the products that have the highest and lowest manufacturing costs

```
SELECT
    f.product_code,
    p.product,
    f.manufacturing_cost
FROM fact_manufacturing_cost f
JOIN dim_product p
ON f.product_code = p.product_code
WHERE manufacturing_cost
IN (SELECT MAX(manufacturing_cost) FROM fact_manufacturing_cost
    UNION
    SELECT MIN(manufacturing_cost) FROM fact_manufacturing_cost
)
ORDER BY manufacturing_cost DESC;
```

Highest Manufacturing Cost



\$240.54

A6120110206
AQ HOME Allin1 Gen 2
Personal Desktop

Lowest Manufacturing Cost



\$0.89

A2118150101
AQ Master wired x1 Ms
Mouse

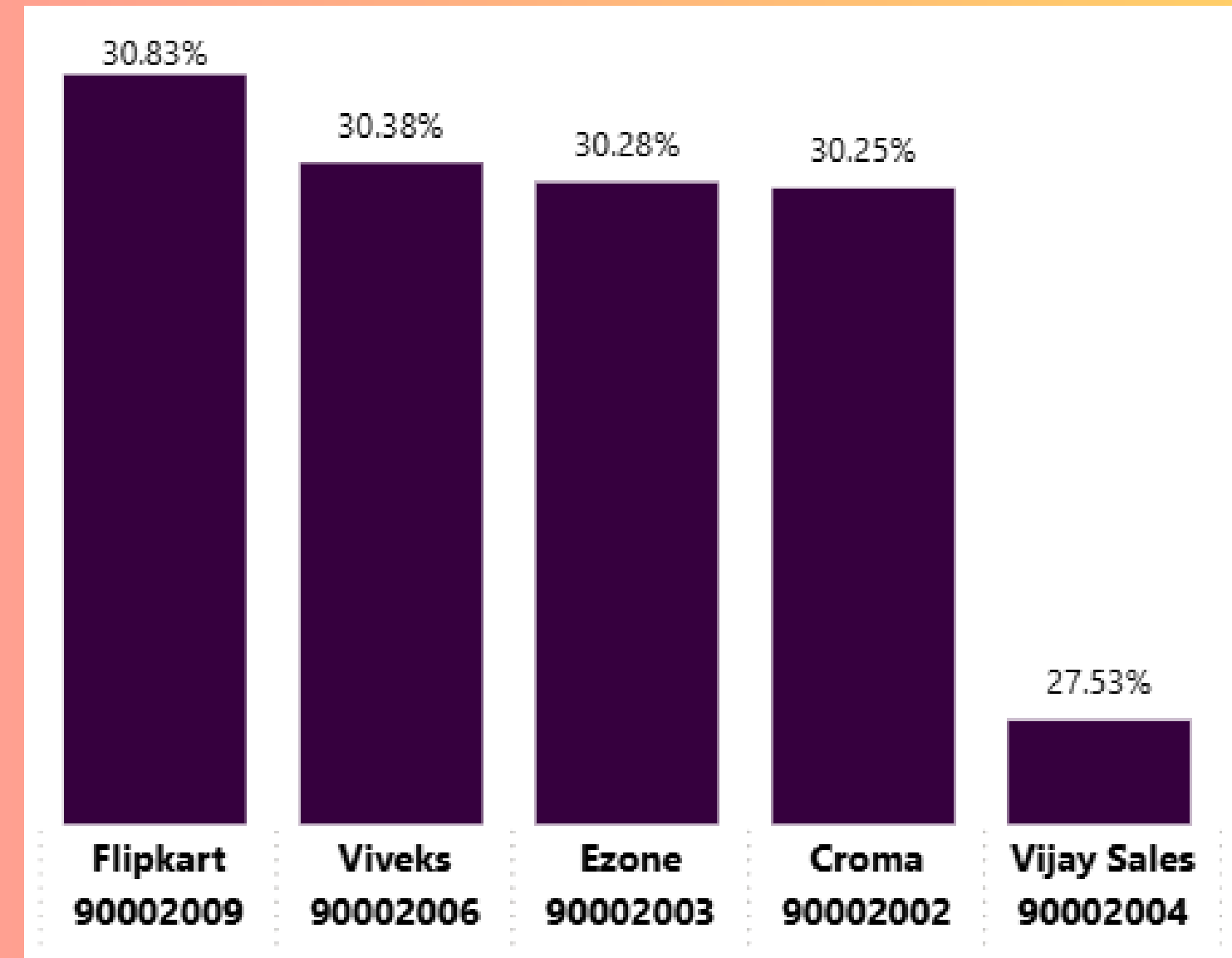
Insights

- Personal Desktop has the Highest Manufacturing Cost of **\$240.54**.
- Mouse has the lowest Manufacturing Cost of **\$0.89**.

Ad Hoc Request - 6

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market

```
SELECT
    c.customer_code,
    c.customer,
    ROUND(AVG(f.pre_invoice_discount_pct), 2)
    AS average_discount_pct
FROM dim_customer c
JOIN fact_pre_invoice_deductions f
ON c.customer_code = f.customer_code
WHERE fiscal_year = 2021 AND
    market = 'India'
GROUP BY c.customer_code, c.customer
ORDER BY average_discount_pct DESC
LIMIT 5;
```

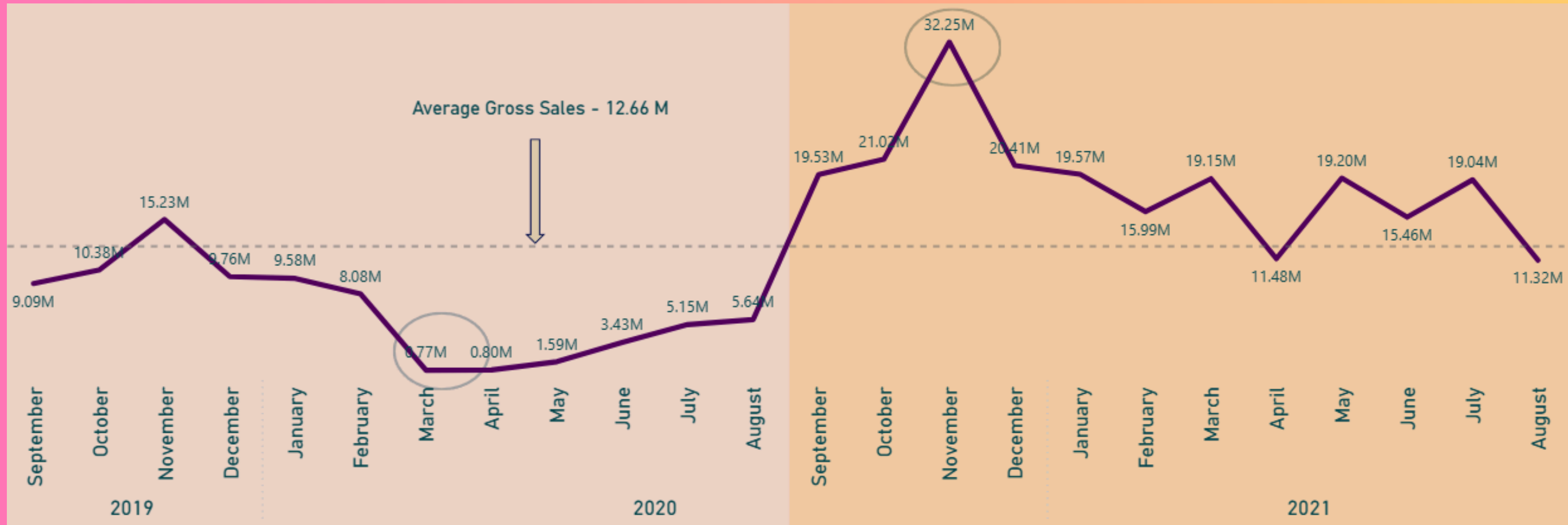


Insights

- Flipkart received the highest Avg. Pre-Invoice-Discount-Pct.

Ad Hoc Request - 7

Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.



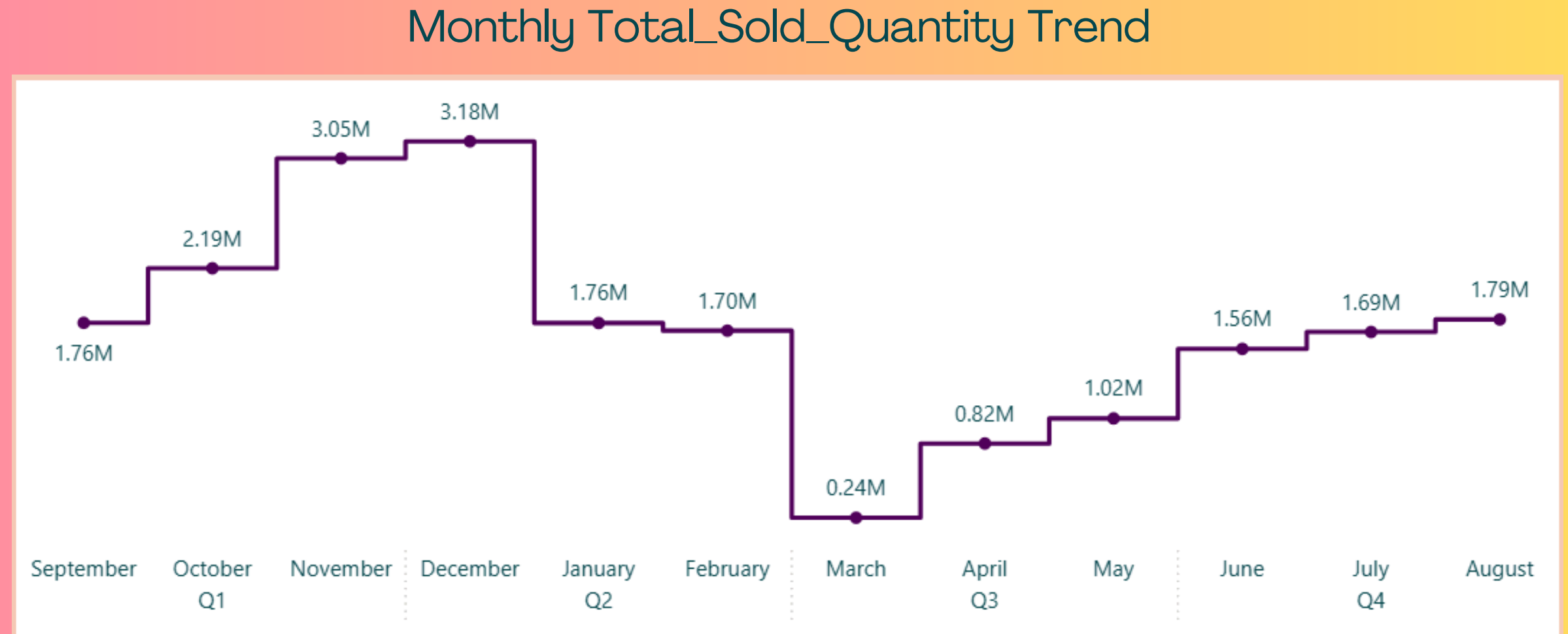
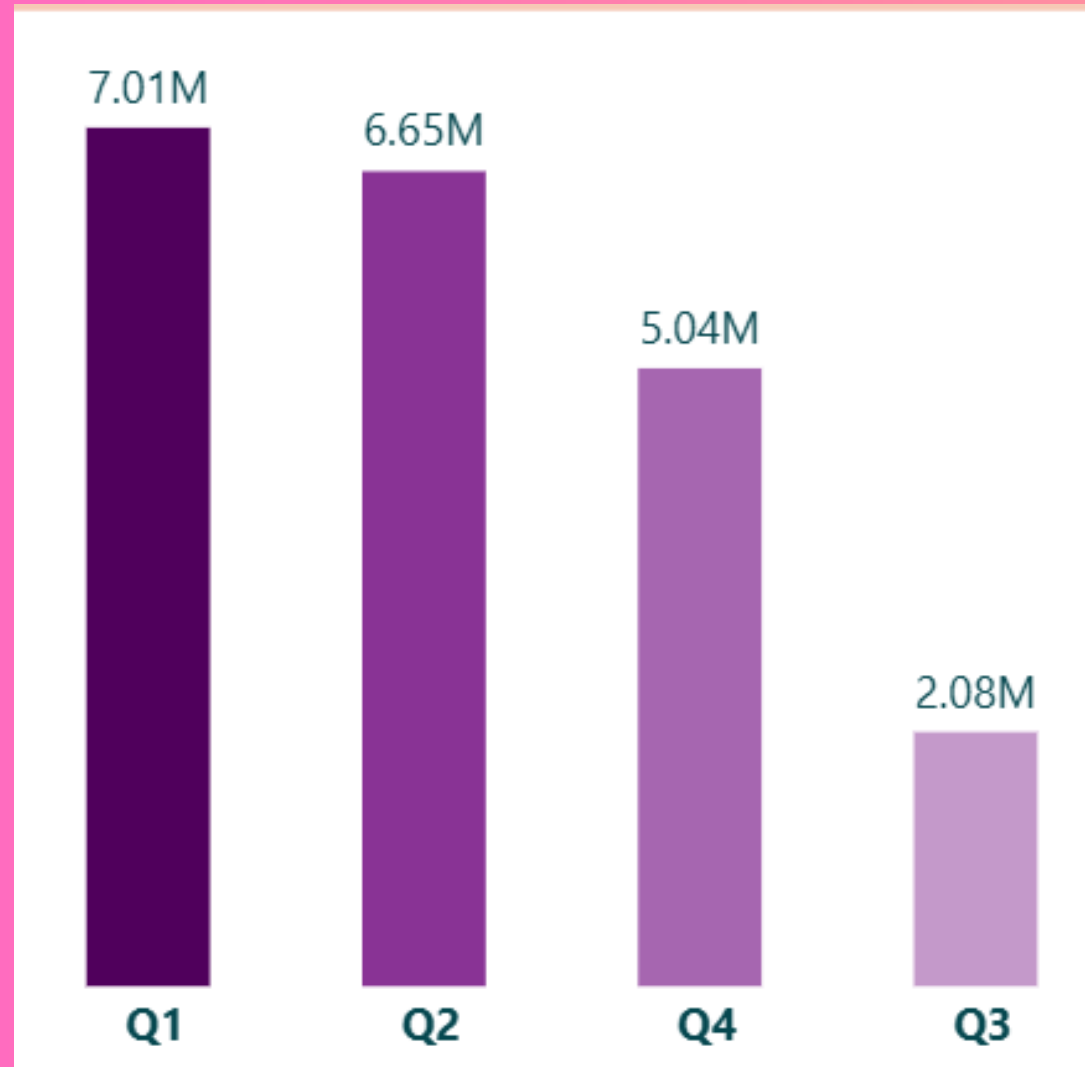
```
SELECT
    MONTHNAME (s.date) AS Month,
    YEAR (s.date) AS Year,
    ROUND((SUM(g.gross_price) * SUM(s.sold_quantity))/1000000, 2)
        AS Gross_sales_Amount_mln
FROM fact_sales_monthly s
JOIN fact_gross_price g
ON s.product_code = g.product_code
AND s.fiscal_year = g.fiscal_year
JOIN dim_customer c
ON s.customer_code = c.customer_code
WHERE c.customer = 'Atliq Exclusive'
GROUP BY Month, Year;
```

Insights

- March - 2020 reported Lowest Gross Sales of \$0.77M
- November - 2020 reported Highest Gross Sales of 32.25M
- Average Gross Sales - 12.66M

Ad Hoc Request - 8

In which quarter of 2020, got the maximum total_sold_quantity?



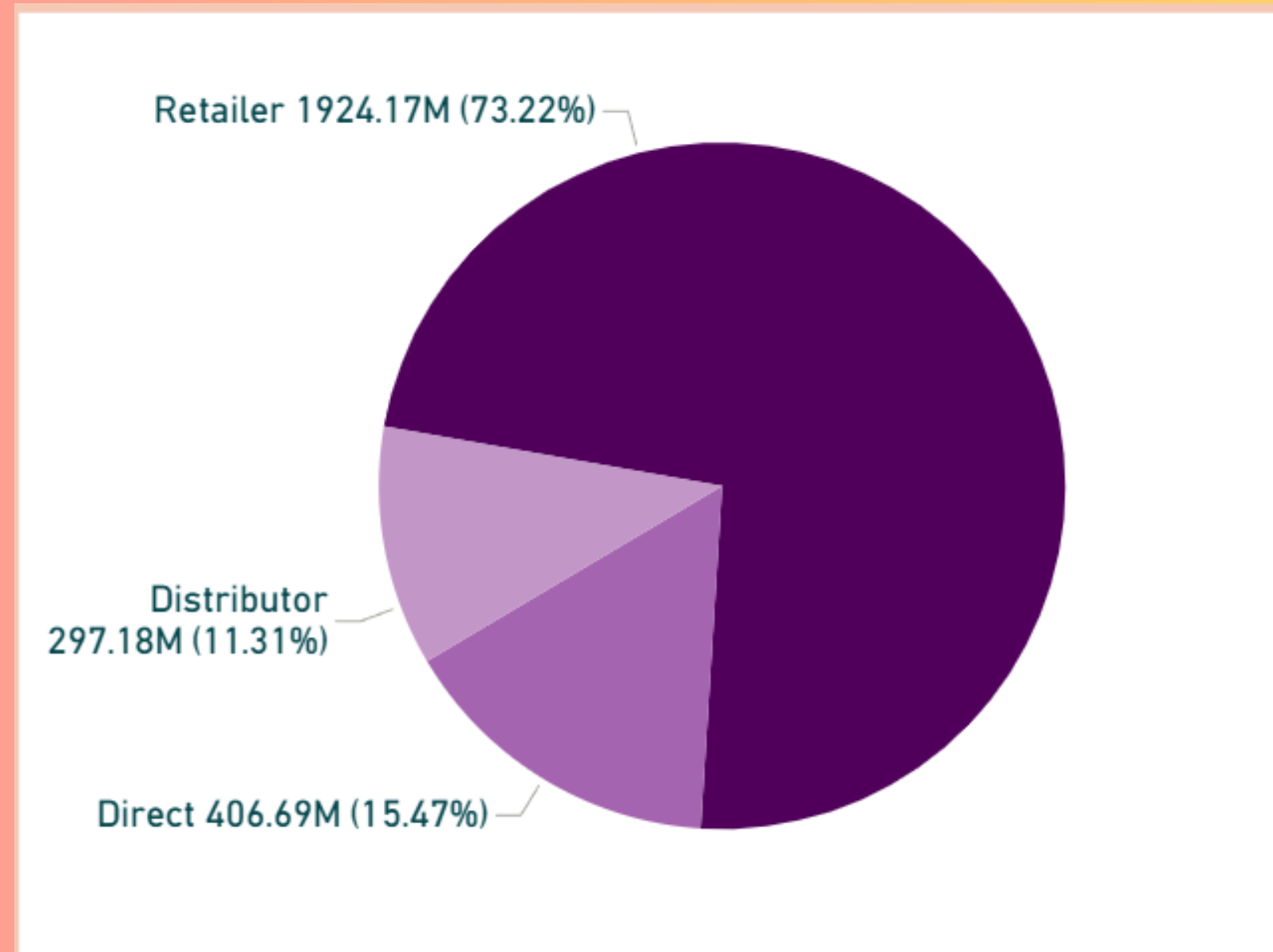
Insights

- In Q1 2020 Company reported the highest Quarterly Sold Quantity of 7.01M followed by Q2 and Q4.
- In monthly December is the highest Quarterly Sold Quantity of 3.18M followed by November (3.05M)
- March month reported lowest monthly total sold quantity.

Ad Hoc Request - 9

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

```
WITH gross_sales AS (  
  SELECT  
    c.channel,  
    g.gross_price * s.sold_quantity AS Gross_sales  
  FROM fact_sales_monthly s  
  JOIN fact_gross_price g  
  ON s.product_code = g.product_code  
  AND s.fiscal_year = g.fiscal_year  
  JOIN dim_customer c  
  ON s.customer_code = c.customer_code  
,  
  
  agg AS (  
    SELECT  
      channel,  
      ROUND(SUM(Gross_sales)/1000000,2) AS Gross_sales_mln  
    FROM gross_sales  
    GROUP BY channel  
  )  
  
  SELECT  
    channel,  
    Gross_sales_mln,  
    ROUND(((Gross_sales_mln / SUM(Gross_sales_mln) OVER()) *100,2) AS percentage  
  FROM agg  
  ORDER BY Gross_sales_mln DESC;
```



Insights

- Retailer channel is contributing more than 73% of total sales in 2021.

Ad Hoc Request - 10

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

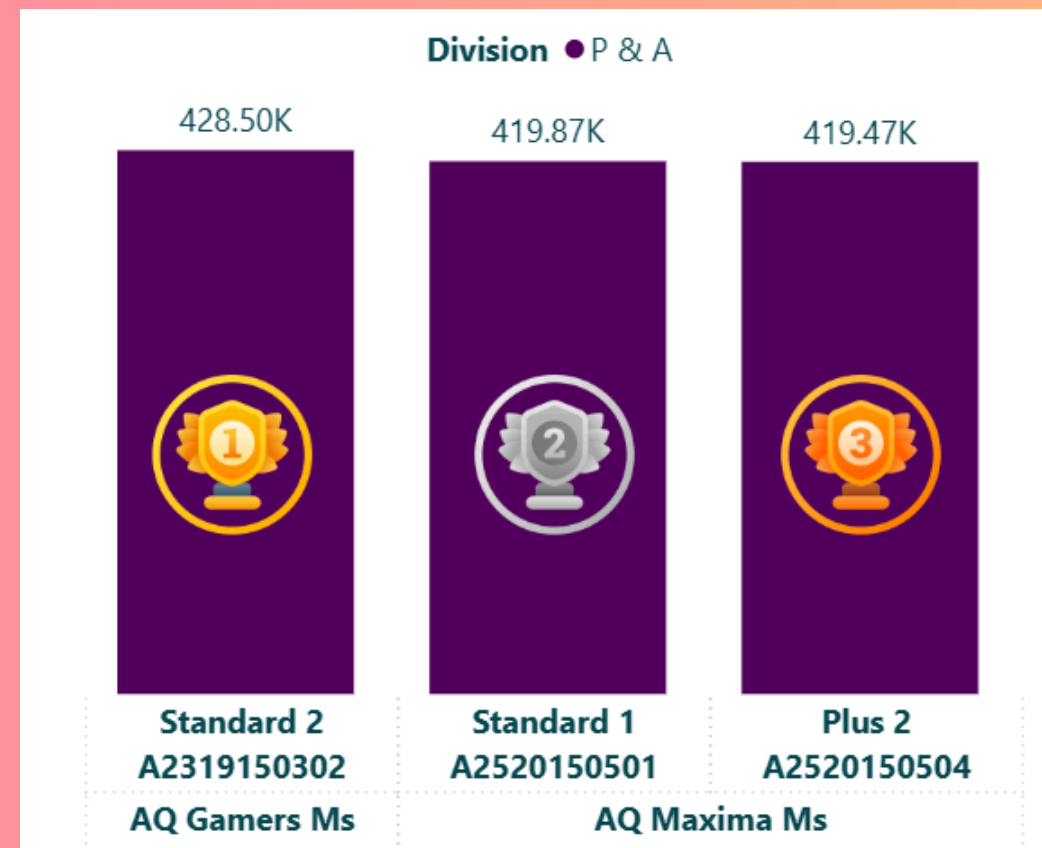
```
WITH CTE1 AS
(
  (SELECT
    p.division,
    p.product_code,
    p.product,
    SUM(f.sold_quantity) AS total_sold_quantity
  FROM dim_product p
  JOIN fact_sales_monthly f
  ON p.product_code = f.product_code
  WHERE fiscal_year = 2021
  GROUP BY p.division, p.product_code, p.product
  ORDER BY total_sold_quantity DESC
  ),
  CTE2 AS
  (
    (SELECT
      *,
      DENSE_RANK()
        OVER(PARTITION BY division
              ORDER BY total_sold_quantity DESC) AS rank_order
    FROM CTE1
    )
  )
SELECT *
FROM CTE2
WHERE rank_order <=3;
```

Top 3 Products in each Divisions

	division	product_code	product	total_sold_quantity	rank_order
▶	N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
	N & S	A6818160202	AQ Pen Drive DRC	688003	2
	N & S	A6819160203	AQ Pen Drive DRC	676245	3
	P & A	A2319150302	AQ Gamers Ms	428498	1
	P & A	A2520150501	AQ Maxima Ms	419865	2
	P & A	A2520150504	AQ Maxima Ms	419471	3
	PC	A4218110202	AQ Digit	17434	1
	PC	A4319110306	AQ Velocity	17280	2
	PC	A4218110208	AQ Digit	17275	3

Ad Hoc Request - 10

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?



Insights

- Top 3 products N&S Division are USB Flash Drives Category.
- Top 3 products P&A Division are Mouse Category.
- Top 3 products PC Division are Personal Laptop Category.

Highlights

- Unique products grew **36.33% YoY (245 → 334)**, signaling strong portfolio expansion.
- Notebook, Accessories & Peripherals dominate **83%** of product portfolio.
- Accessories added **34** new products; Desktop grew **214%** in unique product count.
- Flipkart received highest avg. pre-invoice discount (**30.83%**) in India FY2021.
- Gross Sales report – November 2020 peaked at ₹32.25M; March 2020 was the lowest at ₹0.77M
- Q1 (Sep–Nov) recorded highest sold quantity of **7.01M** units in FY2020.
- Retailer channel drives **73.22% of gross sales** — Direct & Distributor underutilized.
- AQ Pen Drive (N&S), AQ Gamers Ms Mouse (P&A), AQ Digit Laptop (PC) are top products in FY2021

Recommendations

- **Reduce over-dependence on Retailers (73%)** — invest in Direct & Distributor channels to diversify revenue streams and reduce channel risk.
- **Double down on Accessories & Desktop segments** — both showed the strongest product expansion momentum in FY2021 and have room for bundle & loyalty strategies.
- **Audit Flipkart's 30.83% discount** — assess if the volume it drives justifies the margin erosion compared to Viveks (30.38%) and Ezone (30.28%).
- **Build a silicon chip buffer stock** - The March 2020 sales crash to \$0.77M exposed a critical supply chain vulnerability — when chip supply froze, production stopped regardless of demand. A 60-day safety stock policy for key semiconductor components can prevent a repeat zero-sales scenario during future global disruptions.
- **Attack the Q3 slump with pre-season campaigns** - Historical data consistently shows Q3 (March–May) as the weakest quarter, with March recording the lowest monthly sold quantity. Launching targeted promotions — student discounts, bundle deals, and early summer offers — before the dip hits can smooth the revenue curve and lift the quarter's performance.

Thank You

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